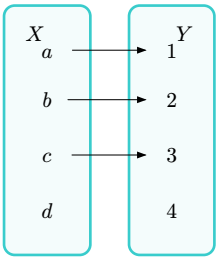
Analytical Methods

- Demand
- Capacity
- Lead Time
- Yield
- Multiple Products
- Bundling
- Discounts



Exercise 1.

$$\begin{aligned} \max \quad & z = (80-20)a + (100-25)b \\ s.t. \quad & 0.2a + 0.5b \leq 60 \\ & a \leq 200 \\ & b \leq 110 \\ & a \geq 0 \\ & b \geq 0 \end{aligned}$$

Constraints

$$0.2a + 0.5b \leq 60$$

$$a = 0 \Rightarrow 0.5b = 60 \Rightarrow b = 120$$

$$b = 0 \Rightarrow 0.2a = 60 \Rightarrow a = 300$$

Objective function

$$60a + 75b$$

$$a = 0 \Rightarrow$$

$$b = 0 \Rightarrow$$

1. Define decision variables
2. Define objective function
3. Define constraints

Assumptions:

- Linearity
- Divisibility (fractional $\neq$ whole number)
- Certainty (parameters non stochastic)
- Non-negativity

Diet Problem

Value (a_i)

	Vitamine	Protein	Calcium	Fat
Corn	8	5	6	8
Beef	2	5	10	8
Soy	6	12	6	4
Fish	8	18	6	5

Bounds (L_j, U_j)

- Vitamine:
  - Min: 60
  - Max: N.A
- Protein:
  - Min: 300

  - Max: N.A
- Calcium:
  - Min: 70
  - Max: N.A
- Fat:
  - Min: 40
  - Max: 200

Availability (s_i):
- Corn: 10
- Beef: 6
- Soy: 4
- Fish: 5

Cost (c_i)
- Corn: 60 c
- Beef: 400c
- Soy 75 c
- Fish: 350 c

Notation

- Ingredients: $i \in I$
- Nutrients: $j \in J$

Decision variables

- x_i: kgs of ingredient i in the meal

$$
\begin{aligned}
\min \quad & \sum_{i \in I} c_i x_i & \\
s.t. \quad & x_i \leq s_i & \forall i \in I \\
& \sum_{i \in I} a_{ij} x_i \geq L_j & \forall j \in J \\
& \sum_{i \in I} a_{ij} x_i \leq U_j & \forall j \in J \\
& x_i \geq 0 & \forall i \in I
\end{aligned}
$$

Scheduling Problem

Requirements

- 12-4: 5
- 4-8: 7
- 8-12: 15
- 12-4: 8
- 4-8: 12
- 8-12: 8

8 Consecutive hours

Minimum number of officers to cover all shifts

Shifts:

- 1: 12-8
- 2: 4-12
- 3: 8-4
- 4: 12-8
- 5: 4-12
- 6: 8-4

x_i: number of employees assigned to shift i

$$
\begin{aligned}
\min \quad & z = \sum_{i=1}^{6} x_i \\
s.t. \quad & x_i \geq s_i \\
& x_1 + x_6 \geq 5 \\
& x_1 + x_2 \geq 7 \\
& x_2 + x_3 \geq 15 \\
& x_3 + x_4 \geq 8 \\
& x_4 + x_5 \geq 12 \\
& x_5 + x_6 \geq 9 \\
& x_i \geq 0, \quad \forall i \in \{1, 2, ..., 6\}
\end{aligned}
$$

Compact Form

- L_j: # of peple needed during time period j ($j = 1, ..., 6$)

$$
L = \begin{pmatrix} 5 \\ 7 \\ 18 \\ 8 \\ 12 \\ 9 \end{pmatrix}
$$

- $T_{ij} = \begin{cases} 1 & \text{if work shift } i \text{ covers period } j \\ 0 & \text{otherwise} \end{cases}$

$$
T = \begin{pmatrix}
0 & 1 & 0 & 0 & 0 & 1 \\
1 & 1 & 0 & 0 & 0 & 0 \\
0 & 1 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 1 & 0 & 0 \\
0 & 0 & 0 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 1 & 1
\end{pmatrix}
$$

$$
\begin{aligned}
\min \quad & z = \sum_{i=1}^{6} x_i \\
s.t. \quad & \sum_{i=1}^{6} T_{ij} \cdot x_i \geq L_j \quad \forall j \in \{1, ..., 6\} \\
& x_i \geq 0, \quad \forall i \in \{1, ..., 6\}
\end{aligned}
$$

Dynamic Pricing

Price Levels (p)	60	54	48	36	25
Demand Multiplier	1	1.5	1.75	2	inf
Weekly Demand (d)	100	150	175	200	inf

- x_i: # of weeks during which I charge a price level i
  - x_{60}: # of weeks during which I charge a price level $\$60$
  - x_{54}: # of weeks during which I charge a price level $\$54$
  - x_{48}: # of weeks during which I charge a price level $\$48$
  - x_{36}: # of weeks during which I charge a price level $\$36$
  - x_s: # of units salvaged
- p_i: Price
- d_i: Demand associated with price i
- D: Total demand (2500)
- W: # of weeks (15)

$$
\begin{aligned}
\max \quad & z = \sum_{i=1}^{4} \underbrace{p_i}_{\text{price}} \cdot \underbrace{x_i \cdot d_i}_{\text{sales}} + \underbrace{x_s \cdot s}_{\text{salvage}} \\
s.t. \quad & \sum_{i=1}^{5} x_i \leq W \\
& \sum_{i=1}^{5} d_i x_i \leq D \\
& x_i \geq 0 \quad \forall i \in \{1, ..., 5\}
\end{aligned}
$$

Network Flow Problems

Transportation

Data:

- Price
- Shipping cost
- Demand
- Route capacity

Decision Variables: units sent from node i to node j

Objective: Max profit / Min Cost

Constraint: Supply, demand, route capacity

	DC1	DC2	DC3
Factory A	4	6	4
Factory B	6	5	2

D.V.: x_{ij}: number of units shipped from plant i to DC j

Params: c_{ij}: unit cost of shipment from i to j

Constraints:

- Supply (S): 70
- Demand (D): 40
- Route capacity (U_{ij}): route capacity on link i to j

$$
\begin{aligned}
\min \quad & z = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij} \\
s.t. \quad & \sum_{j=1}^{n} x_{ij} \leq s_i \quad \forall i \in \{1, ..., m\} \\
& \sum_{i=1}^{m} x_{ij} \geq d_i \quad \forall j \in \{1, ..., n\} \\
& x_{ij} \leq U_{ij} \\
& x_{ij} \geq 0, \quad \forall i \in \{1, 2\} \wedge j \in \{1, 2, 3\}
\end{aligned}
$$

Feasibility check

$S_{\text{total}} > D_{\text{total}}$

Balanced transportation problem:

$$
\sum_{i=1}^{m} s_i = \sum_{j=1}^{n} d_j
$$

Therefore:

$$
\begin{aligned}
\min \quad & z = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij} \\
s.t. \quad & \sum_{j=1}^{n} x_{ij} = s_i \quad \forall i \in \{1, ..., m\} \\
& \sum_{i=1}^{m} x_{ij} = d_i \quad \forall j \in \{1, ..., n\} \\
& x_{ij} \leq U_{ij} \\
& x_{ij} \geq 0, \quad \forall i \in \{1, 2\} \wedge j \in \{1, 2, 3\}
\end{aligned}
$$

Transforming Unbalanced to Balanced: Dummy node

Assignment

Special Case of Transportation Problem

Supply and Demand = 1

Time requirement to setup each machine for completing each job:

	Job 1	Job 2	Job 3	Job 4

Machine 1	14	5	8	7
Machine 2	2	12	6	5
Machine 3	7	8	3	9
Machine 4	2	4	6	10

Decision variable:

$$x_{ij} = \begin{cases} 1 & \text{if machine } i \text{ does job } j \\ 0 & \text{otherwise} \end{cases}$$

- m: number of jobs
- n: number of machines

$$\begin{aligned} \min \quad & z = \sum_{i=1}^{n} \sum_{j=1}^{m} t_{ij} x_{ij} \\ s.t. \quad & \sum_{j=1}^{m} x_{ij} = 1 \quad \forall i \in \{1, ..., n\} \\ & \sum_{i=1}^{n} x_{ij} = 1 \quad \forall j \in \{1, ..., m\} \\ & x_{ij} \geq 0 \quad \forall i \in \{1, ..., n\} \wedge j \in \{1, ..., m\} \end{aligned}$$

Swimmer assignment problem

Each swimmer swims exactly one stroke, and each stroke is swum by exactly one swimmer. We want to minimise the total relay time.

Data

$$t = \begin{pmatrix} 37.7 & 43.3 & 33.3 & 29.2 \\ 32.9 & 33.1 & 28.5 & 26.4 \\ 33.8 & 42.2 & 38.9 & 29.6 \\ 34.4 & 41.8 & 33.6 & 31.1 \end{pmatrix}$$

$$\begin{aligned} \text{rows:} \quad & \text{ca}, \text{ch}, \text{da}, \text{ke} \\ \text{cols:} \quad & \text{ba}, \text{br}, \text{bu}, \text{fr} \end{aligned}$$

Decision variables

$$x_{ij} = \begin{cases} 1 & \text{if swimmer } i \text{ does stroke } j \\ 0 & \text{otherwise} \end{cases}$$

- $i = 1, ..., n$: swimmers ($n = 4$)
- $j = 1, ..., m$: strokes ($m = 4$)

General model

$$\begin{aligned} \min \quad & z = \sum_{i=1}^{n} \sum_{j=1}^{m} t_{ij} x_{ij} \\ s.t. \quad & \sum_{j=1}^{m} x_{ij} = 1 \quad \forall i \in \{1, ..., n\} && \text{(each swimmer does one stroke)} \\ & \sum_{i=1}^{n} x_{ij} = 1 \quad \forall j \in \{1, ..., m\} && \text{(each stroke gets one swimmer)} \\ & x_{ij} \geq 0 \quad \forall i \in \{1, ..., n\}, j \in \{1, ..., m\} \end{aligned}$$

Note: $x_{ij} \in \{0, 1\}$ can be relaxed to $x_{ij} \geq 0$. The constraint matrix is the incidence matrix of a bipartite graph, so it is **totally unimodular**. Every vertex of the feasible region is therefore integral, and the simplex method returns a 0/1 solution. The constraints $x_{ij} \leq 1$ are also implied by the equalities.

Expanded model

$$\begin{aligned}\min \quad z = {} & 37.7x_{\text{ca,ba}} + 43.3x_{\text{ca,br}} + 33.3x_{\text{ca,bu}} + 29.2x_{\text{ca,fr}} \\ & + 32.9x_{\text{ch,ba}} + 33.1x_{\text{ch,br}} + 28.5x_{\text{ch,bu}} + 26.4x_{\text{ch,fr}} \\ & + 33.8x_{\text{da,ba}} + 42.2x_{\text{da,br}} + 38.9x_{\text{da,bu}} + 29.6x_{\text{da,fr}} \\ & + 34.4x_{\text{ke,ba}} + 41.8x_{\text{ke,br}} + 33.6x_{\text{ke,bu}} + 31.1x_{\text{ke,fr}}\end{aligned}$$

$$\begin{aligned}s.t. \quad & x_{\text{ca,ba}} + x_{\text{ca,br}} + x_{\text{ca,bu}} + x_{\text{ca,fr}} = 1 \\ & x_{\text{ch,ba}} + x_{\text{ch,br}} + x_{\text{ch,bu}} + x_{\text{ch,fr}} = 1 \\ & x_{\text{da,ba}} + x_{\text{da,br}} + x_{\text{da,bu}} + x_{\text{da,fr}} = 1 \\ & x_{\text{ke,ba}} + x_{\text{ke,br}} + x_{\text{ke,bu}} + x_{\text{ke,fr}} = 1\end{aligned}$$

$$\begin{aligned}& x_{\text{ca,ba}} + x_{\text{ch,ba}} + x_{\text{da,ba}} + x_{\text{ke,ba}} = 1 \\ & x_{\text{ca,br}} + x_{\text{ch,br}} + x_{\text{da,br}} + x_{\text{ke,br}} = 1 \\ & x_{\text{ca,bu}} + x_{\text{ch,bu}} + x_{\text{da,bu}} + x_{\text{ke,bu}} = 1 \\ & x_{\text{ca,fr}} + x_{\text{ch,fr}} + x_{\text{da,fr}} + x_{\text{ke,fr}} = 1\end{aligned}$$

$$x_{ij} \geq 0 \quad \forall i \in \{\text{ca}, \text{ch}, \text{da}, \text{ke}\}, j \in \{\text{ba}, \text{br}, \text{bu}, \text{fr}\}$$

- $n \cdot m = 16$ variables
- $n + m = 8$ equality constraints, of which only $n + m - 1 = 7$ are independent (the row sums and the column sums both add up to n)

Optimal solution

$$x^* = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad \begin{cases} \text{Carl} & \to \text{freestyle} & (29.2) \\ \text{Chris} & \to \text{breaststroke} & (33.1) \\ \text{David} & \to \text{backstroke} & (33.8) \\ \text{Ken} & \to \text{butterfly} & (33.6) \end{cases}$$

$$z^* = 29.2 + 33.1 + 33.8 + 33.6 = 129.7 \text{ s}$$

Intuition: breaststroke carries the biggest penalty. Chris is about 9 s faster than everyone else at it (33.1 vs ≥ 41.8), so he is placed there, even though freestyle and butterfly are his fastest strokes in absolute terms.

Separable Programming

$$D = \begin{pmatrix} 4 \\ 7 \\ 6 \end{pmatrix}$$

Plant 1	[0, 6] Tons $10	[6, 10] Tons $25
Plant 2	[0, 5] Tons $8	[5, 11] Tons $28

- x_{11}: production quantity at plant 1 at operation level 1 (0 - 6 tons)
- x_{12}: production quantity at plant 1 at operation level 2 (6 - 10 tons)
- x_{21}: production quantity at plant 2 at operation level 1 (0 - 5 tons)
- x_{22}: production quantity at plant 2 at operation level 2 (5 - 11 tons)
- y_{11}: plant 1 → DC 1
- y_{12}: plant 1 → DC 2

…

- c_{ij} cost of shipping from i to j
- x_{ij} production quantity in plant i at level j
- y_{ij} quantity shipped from i to j

$$
\begin{aligned}
\min \quad & 10x_{11} + 25x_{12} + 8x_{21} + 28x_{22} + \sum_{i=1}\sum_{j=1} c_{ij}y_{ij} \\
s.t. \quad & x_{11} + x_{12} + x_{21} + x_{22} = 17 \\
& x_{11} \leq 6 \\
& x_{12} \leq 4 \\
& x_{21} \leq 5 \\
& x_{22} \leq 6 \\
& y_{11} + y_{21} = 4 \\
& y_{12} + y_{22} = 7 \\
& y_{13} + y_{23} = 6 \\
& x_{11} + x_{12} = y_{11} + y_{12} + y_{13} \\
& x_{21} + x_{22} = y_{21} + y_{22} + y_{23} \\
& x_1, ..., x_4 \geq 0
\end{aligned}
$$

Transhipment

Same as **Transportation** problem but with **Transhipment** nodes

- i: supplier node
- j: transhipment node
- k: demand node

D.V.

- x_{ij}: inbound (supplier node $\rightarrow$ transhipment node)
- y_{ij}: outbound (transhipment node $\rightarrow$ demand node)

Constraints

- Demand constraints

$$\sum_j y_{jk} = d_k \quad \forall k \in K$$

- Supply constraints

$$\sum_i x_{jk} = s_i \quad \forall k \in K$$

- Flow conservation constraints

$$\sum_i x_{ij} = \sum_k y_{jk}$$

Montperlier Ski Company

- Initial inventory: 200
- Ending inventory: 1200
- Regular time (50%)
- Overtime (50%)
- Holding cost: 3%

Month	Demand	Capacity	Regular Time Production Cost	Overtime Production Cost
July	400	1000	25	30
August	600	800	26	32
September	1000	400	29	37

- $i \in I$: Month (July, August, September)
- $j \in J$: Production modes (Regular, Overtime)
- x_{ij}: Production on month i at cost j (regular or overtime)

Customers:

- J:
  - Demand: 400
  - Net Demand: 400 - 200 = 200

- A:
  - Demand: 600
  - Net Demand: 600
- S:
  - Demand: 1000
  - Net Demand: 1000 + 1200 = 2200
- Dummy:
  - Demand: Total Supply - Total Demand = 300

Total Demand: 3000

Suppliers:
- JR: 1000
- JO: 500
- AR: 800
- AO: 400
- SR: 400
- SO: 200

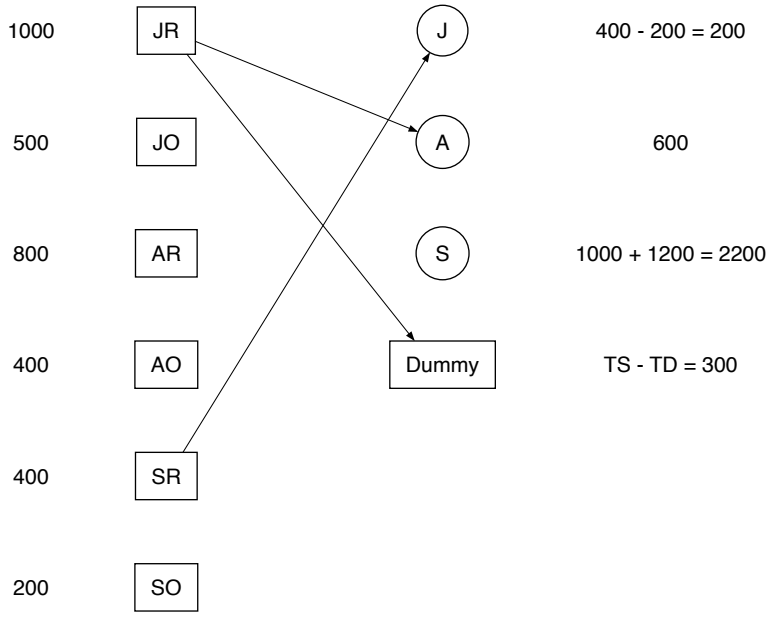
Total Supply: 3300

$$
\begin{aligned}
\min \quad & \sum_{i \in I} \sum_{j \in J} x_{ij} c_{ij} \\
s.t. \quad & \sum_{i} x_{ij} = (\vee \leq) d_j, \quad \forall j \in J \\
& \sum_{j} x_{ij} = (\vee \leq) s_j, \quad \forall i \in I \\
& x_{ij} \geq 0, \quad \forall i \in I \wedge j \in J
\end{aligned}
$$

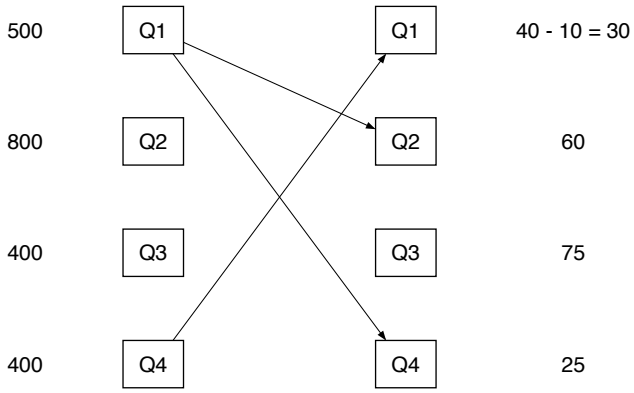
$$
c_{13} = 25 + (3\% \times 24) \times 2
$$

Cost + 3% Cost for 2 months (produce in July sold in September)

Making Produced in september sold in july impossible (use M)



Sailco Inventory
- Regular capacity: 40
- Regular cost: 400
- Overtime capacity: infinity
- Overtime cost: 450
- h: 20
- Beginning inventory: 10



Since holding cost is constant (unlike Montperlier Ski Company example), we don't need transportation formulation.

- x_t: number of boars producted in quarter t using regular time
- y_t: number of boars producted in quarter t using overtime
- i_t: ending inventory for quarter t

$$
\begin{aligned}
\min \quad & \sum_{t=1}^{4}(400x_t + 450y_t) + \sum_{t=1}^{4} 20 \cdot i_t \quad \text{production cost + holding cost} \\
s.t. \quad & x_t \leq 40 \quad \forall t = 1, ..., 4 \\
& i_t = i_{t-1} + x_t + y_t - d_t \quad \forall t = 1, ..., 4 \\
& x_t, y_t \geq 0 \quad \forall t = 1, ..., 4 \\
& i_t \geq 0 \quad \forall i = 1, ..., 4 \quad \text{no backorders}
\end{aligned}
$$

Adding backorder penalty:

$b = 100$

2 new variables:

$$
\begin{aligned}
i_t^+ &= \#\text{ onhand} \\
i_t^- &= \#\text{ backorders} \\
i_t &= i_t^+ - i_t^-
\end{aligned}
$$

$$
\begin{aligned}
\min \quad & \sum_{t=1}^{4}(400x_t + 450y_t) + \sum_{t=1}^{4} 20 \cdot i_t \quad \text{production cost + holding cost} \\
s.t. \quad & x_t \leq 40 \quad \forall t = 1, ..., 4 \\
& i_t^+ - i_t^- = i_{t-1}^+ - i_{t-1}^- + x_t + y_t - d_t \quad \forall t = 1, ..., 4 \\
& x_t, y_t \geq 0 \quad \forall t = 1, ..., 4 \\
& i_t^+ \geq 0 \quad \forall i = 1, ..., 4 \\
& i_t^- \geq 0 \quad \forall i = 1, ..., 4
\end{aligned}
$$

Product Mix

$$
\text{bom} = \begin{pmatrix} 2 & 1 \\ 2 & 2 \end{pmatrix}
$$

$$
\text{capacity} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}
$$

$$
\text{price} = \begin{pmatrix} 20.00 \\ 15.00 \end{pmatrix}
$$

```
import gurobipy as gp
from gurobipy import GRB
import numpy as np

products = ["table", "chair"]
bom = np.array([[2, 1],
                [2, 2]])
capacity = np.array([6, 8])
price = np.array([20.00, 15.00])

m = gp.Model("product_mix")
x = m.addMVar(shape=price.size, lb=0.0, name=products)
```

```
m.setObjective(price @ x, GRB.MAXIMIZE)
cap = m.addConstr(bom @ x <= capacity, name="capacity")

m.optimize()

if m.Status == GRB.OPTIMAL:
    obj = x.Obj
    inc = x.SAObjUp - obj
    dec = obj - x.SAObjLow

    print(f"Objective = {m.ObjVal:.2f}\n")

    print(f"{'Var':6} {'Value':>7} {'Obj':>7} {'Inc':>7} {'Dec':>7} {'RC':>7}")
    for row in zip(x.VarName, x.X, obj, inc, dec, x.RC):
        print(f"{row[0]:6} " + " ".join(f"{v:7.2f}" for v in row[1:]))

    print(f"\n{'Constr':12} {'Slack':>7} {'Dual':>7} {'RHS lo':>7} {'RHS hi':>7}")
    for row in zip(cap.ConstrName, cap.Slack, cap.Pi, cap.SARHSLow, cap.SARHSUp):
        print(f"{row[0]:12} " + " ".join(f"{v:7.2f}" for v in row[1:]))
```

Sensitivity Analysis

- How does the solution change
  - If the RHS of constraint change?
  - If an objective coefficient changes?
- What is the range of optimality and feasibility?
  - Allowable change in the objective function coefficient
  - Allowable change in the RHS

Ice Cream

- Check allowable Increase / Decrease
- If in range:
  - New Obj Value = New Obj Value + (Decrease) $\times$ Shadow Price

100% rule

- Simultaneous change in objective function
- Simultaneous change in coefficients

abs(New value - Old value) / allowable value + abs(New value - Old value) / allowable value <= 1 (100%)

if >= 100% cannot use shadow price to calculate new profit

Integer Programming

Knapsack Problem

TBA Airlines

$$\begin{aligned} \max \quad & S + 5L \\ s.t. \quad & 5S + 50L \\ & S \leq 2 \\ & S, L \geq 0 \\ & S, L \in \{0, 1\} \end{aligned}$$

$$\begin{aligned} \text{LP} &\geq \text{IP} \quad (\max) \\ \text{LP} &\leq \text{IP} \quad (\min) \end{aligned}$$

Distribution Channel Restructuring

Warehouse ($m = 5$):

SEA, DEN, STL, ATL, PHI

Market ($n = 6$):

NW, SW, UMW, LMW, NE, SE

Operating cost:

$$c_o = ()$$

Capacity:

$$k = ()$$

Demand:

$$D = ()$$

Transportation cost:

$$c_T = ()$$

Decision variables:

- x_{ij}: # of units shipped from WH i to customer j
- c_{ij}: unit cost of a shipment from WH i to customer j
- s_i: capacity of WG i
- y_i: open or close WH i

$$\begin{cases} 1 & \text{if WH } i \text{ open} \\ 0 & \text{otherwise} \end{cases}$$

- k_i: operational cost of WH i

$$\begin{aligned} \min \quad & \sum_{i=1}^{5}\sum_{j=1}^{6} c_{ij}x_{ij} + \sum_{i=1}^{5} k_i y_i \\ s.t. \quad & \sum_{j=1}^{6} x_{ij} \le s_i y_i, \quad \forall i = 1, ..., 5 \\ & \sum_{i=1}^{5} x_{ij} \ge d_j, \quad \forall j = 1, ..., 6 \\ & \sum_{j=1}^{6} x_{ij} \le s_i y_i \quad \forall i = 1, ..., 5 \\ & x_{ij} \ge 0 \quad \forall i, j \\ & y_i \in \{0, 1\} \end{aligned}$$

Capital Budgeting

$$y_i = \begin{cases} 1 & \text{if project } i \text{ is chosen} \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} \max \quad & 35y_1 + 18y_2 + 24y_3 + 16y_4 \\ s.t. \quad & 10y_1 + 8y_2 + 6y_3 + 12y_4 \le 30 \\ & 8y_1 + 5y_2 + 4y_3 + 0y_4 \le 15 \\ & 8y_1 + 0y_2 + 6y_3 + 0y_4 \le 12 \\ & y_i \in \{0, 1\} \quad \forall i \end{aligned}$$

If budget saved in one year can be used in subsequent years:

$$\begin{aligned} \max \quad & 35y_1 + 18y_2 + 24y_3 + 16y_4 \\ s.t. \quad & 10y_1 + 8y_2 + 6y_3 + 12y_4 \le 30 \\ & 8y_1 + 5y_2 + 4y_3 + 0y_4 \le 15 + (30 - 10y_1 + 8y_2 + 6y_3 + 12y_4) \\ & 8y_1 + 0y_2 + 6y_3 + 0y_4 \le 12 + (15 + 30 - 10y_1 + 8y_2 + 6y_3 + 12y_4) - (8y_1 + 5y_2 + 4y_3 + 0y_4) \\ & y_i \in \{0, 1\} \quad \forall i \end{aligned}$$

At least on project:

$$y_1 + y_2 + y_3 \ge 1$$

Sequence (4 needs to be done before 3):

$$y_3 \le y_4$$

Either 1 or 2, but not both:

$$y_1 + y_2 = 1$$

No more than 2 projects:

$$y_1 + y_2 + y_3 + y_4 \leq 2$$

Set Covering Problem

$$y_i = \begin{cases} 1 & \text{if team in country } i \\ 0 & \text{otherwise} \end{cases}$$

Adjacency matrix:

$$A = ()$$

$$a_{ij} = \begin{cases} 1 \text{ if county } i \text{ is adjacent to county } j \\ 0 \text{ otherwise} \end{cases}$$

$$\begin{aligned} \min \quad & \sum_{i=1}^{37} y_i \\ s.t. \quad & \sum_{j=1}^{37} a_{ij} y_i \geq 1 \quad \forall i = 1, ..., 37 \\ & y_i \in \{0, 1\} \quad \forall i, ..., 37 \end{aligned}$$

Olympic Gymnastic Team
Sets

Indices

Parameters

- B:
- F:

Decision variables

- $x_i = \begin{cases} 1 \text{ if gymnast } i \text{ does BB only} \\ 0 \text{ otherwise} \end{cases}$
- $y_i = \begin{cases} 1 \text{ if gymnast } i \text{ does FE only} \\ 0 \text{ otherwise} \end{cases}$
- $w_i = \begin{cases} 1 \text{ if gymnast } i \text{ does both} \\ 0 \text{ otherwise} \end{cases}$
- bb_i: score of gymnist i for BB
- fe_i: score of gymnist i for FE

Model

$$\begin{aligned} \max \quad & \sum_{i \in} ([x_i + w_i]\text{bb}_i + [y_i + w_i]\text{fe}_i) \\ s.t. \quad & \sum_{i \in} w_i = 3 \\ & \sum_{i \in} (x_i + w_i) = 4 \\ & \sum_{i \in} (y_i + w_i) = 4 \\ & x_i + y_i + w_i \leq 1 \quad \forall i \in \\ & x_i, y_i, w_i \geq 0 \end{aligned}$$

Euing Gas

- x: amount of oil 1 (extra) to be purchased
- x_{ij}: amount of oil i used to product gas j

$$\begin{aligned} & \max \\ s.t. \quad & \end{aligned}$$

Scheduling Problem

- $x_{ik} = \begin{cases} 1 \text{ if customer } i \text{ is assigned to position } k \\ 0 \text{ otherwise} \end{cases}$
- t_k: tardiness of the kth job in sequence
- d: due date

- p: processing time

$$
\begin{aligned}
\min \quad & z = \sum_{k \in K} t_k \\
s.t. \quad & \sum_{k \in K} x_{ik} = 1 && \forall i \in I \\
& \sum_{i \in I} x_{ik} = 1 && \forall k \in K \\
& \sum_{i \in I} p_i \sum_{u}^{k} x_{iu} - \sum_{i \in I} d_i x_{ik} \leq t_k && \forall k \\
& t_k \geq 0 \\
& x_{ik} = \{1, 0\}
\end{aligned}
$$

Cumulative sum p of the previous customer orders minus the due date (d) of the current task

$$t_1 = (p_3 - d_3)^+$$

Due date of the kth job in the sequence:

$$\sum_{i \in I} x_{ik} d_i$$

Completion time:

$$\sum_{i \in I} p_i \sum_{u}^{k} x_{iu}$$

This is non linear:

$$t_k = \left(\sum_{i \in I} p_i \sum_{u}^{k} x_{iu} - \sum_{i \in I} x_{ik} d_i \right)^+$$

So we replace it with:

$$
\begin{aligned}
& \sum_{i \in I} p_i \sum_{u}^{k} x_{iu} - \sum_{i \in I} d_i x_{ik} \leq t_k \\
& t_k \geq 0
\end{aligned}
$$

Sequence-dependent setup time production problem
TSP Problem

- $x_{ik} = \begin{cases} 1 \text{ if gasoline } i \text{ is assigned to position } k \\ 0 \text{ otherwise} \end{cases}$
- t_{ij}: transition time between gasoline i and j

$$
\begin{aligned}
\min \quad & \sum_{k \in K} t_k \\
s.t. \quad &
\end{aligned}
$$

Solution:

- LFR → LFP → LP → LR → LFR
- 330 minutes

TSP

- $x_{ij} = \begin{cases} 1 \text{ if vertex } j \text{ is visited immediately after vertex } i \\ 0 \text{ otherwise} \end{cases}$

$$
\begin{aligned}
\min \quad & \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij} \\
s.t. \quad & \sum_{i} x_{ij} = 1 && \forall j && \text{(in)} \\
& \sum_{i} x_{ij} = 1 && \forall i && \text{(out)} \\
& \sum_{(i,j) \in S} x_{ij} \leq |S| - 1 && \forall S \subset V, |S| > 1 && \text{(prevent subtours)}
\end{aligned}
$$

Alternative:

$$
\begin{aligned}
\min \quad & \sum_{i\in I}\sum_{j\in J} c_{ij}x_{ij} & & \\
s.t. \quad & \sum_{i} x_{ij} = 1 & \forall j \quad & \text{(in)} \\
& \sum_{i} x_{ij} = 1 & \forall i \quad & \text{(out)} \\
& u_1 = 1 & & \\
& 2 \le u_i \le n & \forall i \ge 2 & \\
& u_i - u_j + 1 \le (n-1)\big(1 - x_{ij}\big) & \forall i, j \ne 1 &
\end{aligned}
$$

Shipment Consolidation & Dispatching

- k: order ($k \in K$)
- i_k: destination of order k
- w_k: weight of order k
- r_k: release time of order k (day order k is ready for delivery)
- d_k: deadline for order k
- τ_{kr}:
- v_k:
- S_r:

2 delivery options:

- FTL
- LTL

Decision Variables

- $x_{krt} = \begin{cases} 1 \\ 0 \end{cases}$
- $y_{rt} = \begin{cases} 1 \\ 0 \end{cases}$
- $v_k = \begin{cases} 1 \\ 0 \end{cases}$

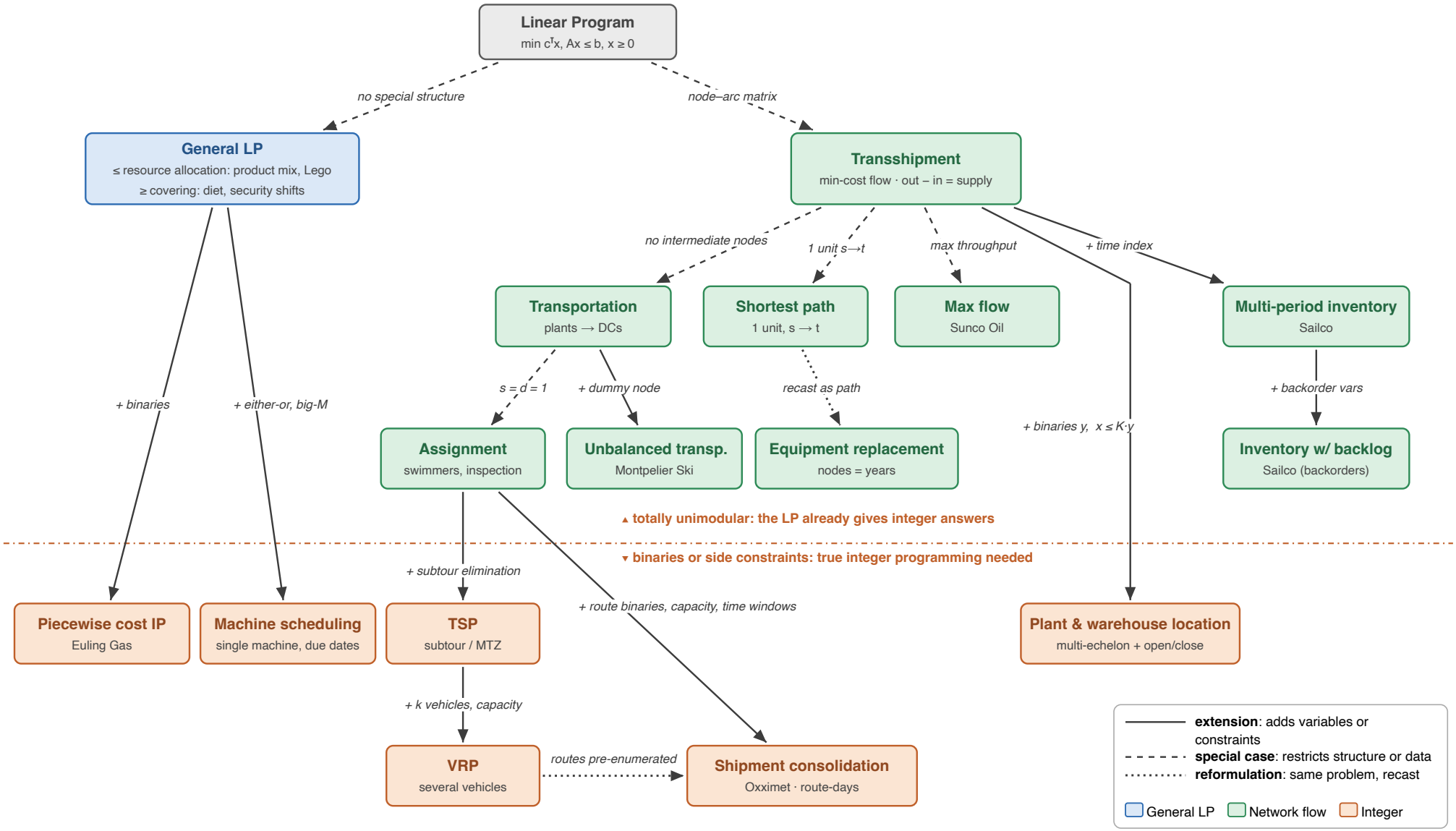
v_k is defined only if $r_k + \tau'_k \le d_k$

$$
\begin{aligned}
\min \quad & \sum_{r\in R}\sum_{t\in T} f_r t_{rt} + \sum_{k\in K} g_k v_k & & \\
\text{s.t.} \quad & \sum_{k: r_k \le t \le d_k - \tau_{kr}, i_k \in S_r} w_k x_{krt} \le q_r y_{rt} & & r \in R, t \in T \\
& \sum_{r: i_k \in S_r} \sum_{t: r_k \le t \le d_k - \tau_{kr}} x_{krt} + v_k = 1 & & k \in K \\
& x_{krt} \in \{0,1\} & & k \in K, r \in R, t \in T \\
& y_{rt} \in \{0,1\} & & r \in R, t \in T \\
& v_k \in \{0,1\}, & & k \in K
\end{aligned}
$$

$$
R = \begin{pmatrix} 800 \\ 750 \\ 400 \\ 780 \\ 830 \\ 830 \\ 860 \end{pmatrix}
$$

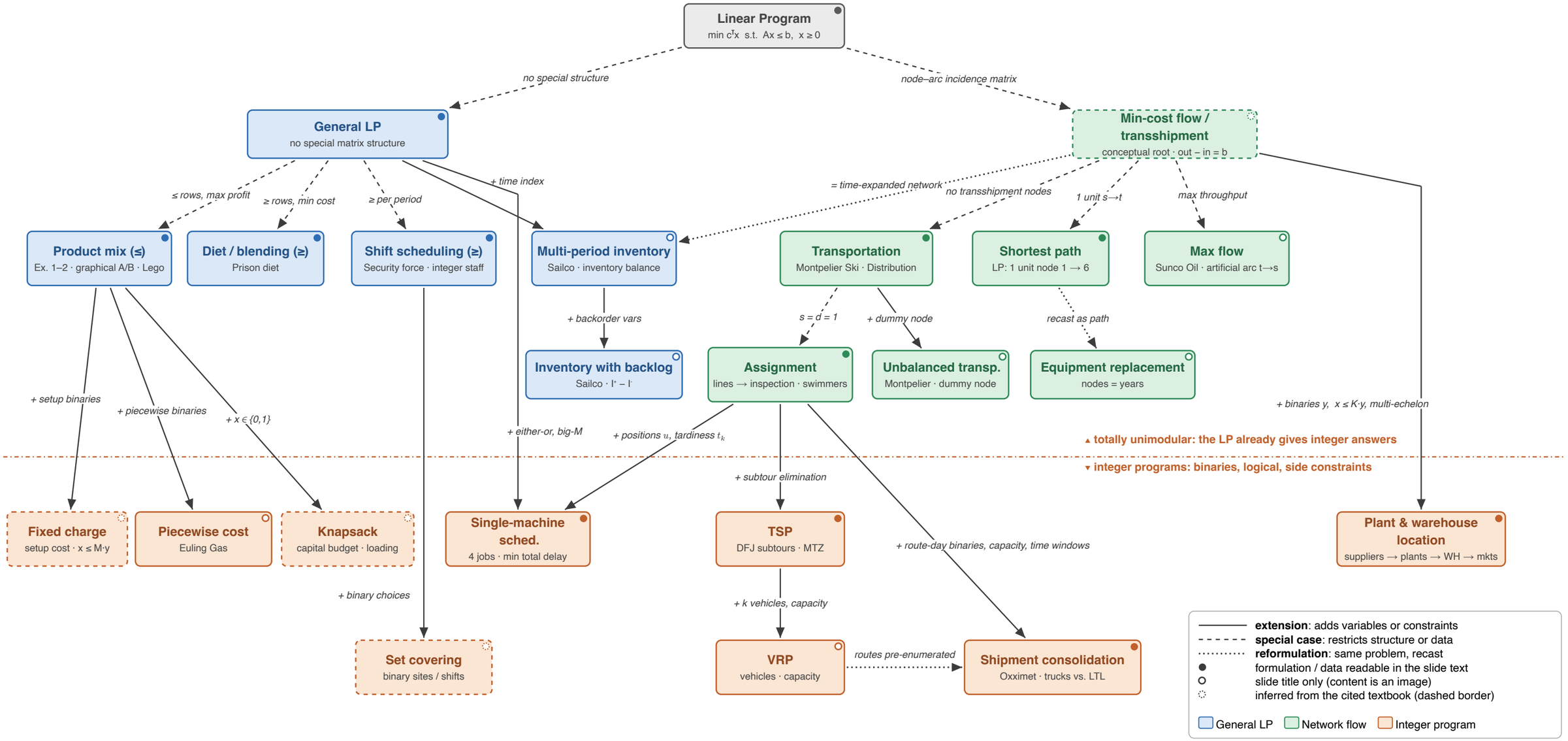
Taxonomy of optimization models

Arrows run from simpler to richer models. A model can have several parents, so this is a DAG rather than a tree.



Linear & integer programming models in MZ 510

Arrows run from simpler to richer models. A model can have several parents, so this is a DAG, not a tree.



Catalogue: every LP / IP item in the slides ● readable ○ title only ◌ inferred

#	Model / example	Graph node	What it adds / key feature	Ev.
LP Part 1: formulation & graphical solution				
1	Exercise 1: products A, B on one bottleneck machine	Product mix	60 h machine (≤), demand caps $x \le D$	●
2	Exercise 2: products A, B, C, machine + labour hours	Product mix	second resource row (10 labour h)	●
3	Graphical solution: A/B with labour, parts, freight	Product mix	feasible region, isoprofit line, corner point $Z^* = 52{,}000$ at (100, 40)	●
4	Introductory formulation slides (Dr. J. Xu)	—	content is images; likely LP terminology and assumptions	◌
LP Part 2: network & multi-period models				
5	Assignment: 5 assembly lines → 5 inspection areas	Assignment	network representation, $S_i = D_j = 1$	●
6	Montpelier Ski: balanced transportation (formulation + solution)	Transportation	supply = demand	○
7	Montpelier Ski: unbalanced transportation	Unbalanced transp.	dummy demand/supply point or ≤ supply rows	○
8	Sailco inventory, no backlogging (Winston Ch. 3)	Multi-period inventory	$I_t = I_{t-1} + x_t - d_t$ regular vs. overtime production	○
9	Sailco with backlogging	Inventory with backlog	$I_t = I^+_t - I^-_t$, backorder cost	○
10	"Formulate an LP and check your solution"	—	exercise; content not readable	○
11	Sunco Oil maximum flow (+ alternative without artificial arc)	Max flow	arc capacities; max flow on artificial arc t → s, or max net outflow of s	○
12	Shortest path LP (Dr. J. Goentzel): 6 nodes, 7 arcs	Shortest path	flow conservation (+1 / 0 / –1); $x_j \in \{0,1\}$ relaxes to $x_j \ge 0$	●
13	Equipment replacement (3 slides)	Equipment replacement	nodes = years, arc (i, j) = buy at i, sell at j	○
LP Part 3: Excel Solver & sensitivity				
14	Lego production: tables & chairs	Product mix	bill of materials, large/small bricks	●
15	Prison diet: milk, beans, oranges	Diet / blending	min cost, ≥ niacin, thiamin, vitamin C	●
16	Security force scheduling: 6 overlapping 8-h shifts	Shift scheduling	each 4-h period ≥ requirement; integer officers (32)	●
17	Distribution problem: 2 plants → 3 DCs	Transportation	supply ≤, demand ≥, cost $450	●
18	Swimmers → medley-relay strokes	Assignment	unbalanced: 5 swimmers ≤ 1, 4 strokes = 1	●
19	Sensitivity report	(method)	final value, reduced cost, shadow price, allowable ± ; range of optimality / feasibility	●
Integer Programming				
20	IP introduction (Dr. P. Yadav), 30 image slides	Fixed charge · Knapsack · Set covering	likely: binaries, capital budgeting, fixed charge, set covering, either-or / if-then, LP relaxation, branch & bound	◌
21	Truck-loading image ("loads")	Knapsack	loading/packing motivation	◌
22	Euling Gas (Winston Ch. 9)	Piecewise cost	"c(x) is nonlinear" → transformed into an IP	○
23	Plant & warehouse location	Plant & warehouse location	binaries y_i, y_e supplier, plant & WH capacities; flow balance at plants & WH	●
24	Single-machine scheduling with due dates (Winston Ch. 9)	Single-machine sched.	4 jobs, min total delay; either-or (big-M) or positional formulation	●
25	Shipment consolidation & dispatching (Ghiani et al.): Oxximet	Shipment consolidation	route-day binaries y_{rt}, x_{krt} LTL option v_k capacity q_r release r_k / deadline d_k	●
26	Travelling salesman (Paolucci)	TSP	degree constraints + subtour elimination (DFJ) or MTZ u_i	●
27	Vehicle routing problem (2 title slides + images)	VRP	multiple vehicles, capacity	○

Modelling devices: the “added constraints and variables”

Device	Typical form	Where it appears	Ev.
Continuous LP devices			
Resource capacity	$\sum_j a_{ij}x_j \le b_i$	product mix, Lego	●
Demand bound	$x_j \le D_j$	Exercises 1–2	●
Covering requirement	$\sum_j a_{ij}x_j \ge r_i$	diet, security shifts	●
Supply / demand rows	$\sum_j x_{ij} \le s_i$, $\sum_i x_{ij} \ge d_j$	transportation, distribution	●
Dummy node	extra column/row with zero cost absorbs imbalance	unbalanced Montpelier	○
Flow conservation	$\sum_{\text{out}} x - \sum_{\text{in}} x = b_v$ (+1 / 0 / –1)	shortest path, max flow, location	●
Arc capacity	$0 \le x_{ij} \le u_{ij}$	Sunco Oil	○
Artificial return arc	arc $t \to s$, maximise its flow	Sunco Oil	○
Inventory balance	$I_t = I_{t-1} + x_t - d_t$	Sailco	○
Backlogging	$I_t = I_t^+ - I_t^-$, cost on I_t^-	Sailco with backlog	○
Integer devices			
Integrality	$x \in \mathbb{Z}_{\ge 0}$	security officers, IP section	●
Binary decision	$y \in \{0,1\}$: open / select / do	location, routes, scheduling	●
Linking (fixed charge)	$x \le My$ or $x \le K_i y_i$	plant & warehouse location	●
Piecewise-linear cost	segment binaries, or $x = \sum \lambda_k b_k$	Euling Gas	○
Either-or (disjunctive)	$S_i + p_i \le S_j + M(1 - y_{ij})$, and the reverse	single-machine scheduling	●
If-then / k-of-n	$g(x) \le My$, $f(x) \le M(1-y)$; $\sum y \le k$	IP introduction	◌
Assignment rows	$\sum_j x_{ij} = 1$, $\sum_i x_{ij} = 1$	assignment, TSP, positional scheduling	●
Positional completion time	$\sum_i p_i \sum_{u=1}^{k} x_{iu}$	positional scheduling	●
Tardiness linearisation	$t_k \ge C_k - d_k$, $t_k \ge 0$ (= max(0, ·))	single-machine scheduling	●
Subtour elimination (DFJ)	$\sum_{i,j \in S} x_{ij} \le \lvert S\rvert - 1$, for all $S \subset V$, $2 \le \lvert S\rvert \le n-1$	TSP	●
MTZ ordering	$u_i - u_j + n x_{ij} \le n - 1$, $2 \le u_i \le n$	TSP	●
Vehicle / route capacity	$\sum_k w_k x_{krt} \le q_r y_{rt}$	consolidation, VRP	●
Time windows	order k may use day t only if $r_k \le t$ and $t + \tau_{kr} \le d_k$	shipment consolidation	●
Mode choice	$\sum_{r,t} x_{krt} + v_k = 1$	consolidation (truck vs. LTL)	●
Pre-enumerated routes	variables per route $r \in R$ instead of per arc	consolidation	●

Base structure × modifier

★ = in the slides · grey = valid combination you can formulate yourself · — = adds nothing new

	Time index x_t, balance across periods	**Binaries / fixed charge** $y \in \{0,1\}$, $x \le M \cdot y$	**Logical (big-M)** either-or, if-then	**Piecewise / nonlinear** segment binaries	**Subtour elimination** DFJ cuts or MTZ u_i
General LP ≤ / ≥ rows	★ Sailco (± backlog)	★ Knapsack / capital budget (inferred) ★ Fixed charge (inferred)	★ Single-machine scheduling (either-or)	★ Euling Gas	—
Covering ≥ rows	*Multi-week rostering*	★ Set covering (inferred)	*Min-shift-length rules*	*Overtime cost tiers*	—
Transportation bipartite supply ⟶ demand	*Production planning as transportation*	*Fixed-charge transportation*	*Min-shipment-or-zero*	*Quantity-discount freight*	—
Assignment s = d = 1	★ Positional machine scheduling ★ Shipment consolidation	already integral (TU)	*Conflict / precedence pairs*	—	★ TSP ⟶ VRP
Transshipment min-cost flow	★ Sailco as time-expanded network	★ Plant & warehouse location	*Single-sourcing per customer*	*Concave arc costs*	—
Shortest path 1 unit s ⟶ t	★ Equipment replacement	already integral (TU)	*Resource-constrained SP*	—	—
Max flow max throughput	*Dynamic (time-expanded) flow*	*Network interdiction*	—	—	—

Solution methods covered alongside the models

Method	**Use**	**Ev.**
Graphical solution	2-variable LPs: feasible region, isoprofit line, optimal corner point	●
Excel Solver	Spreadsheet models of product mix, diet, scheduling, transportation, assignment	●
Sensitivity analysis	Shadow prices, reduced costs, range of optimality (cost coeff.), range of feasibility (RHS)	●
Total unimodularity	Why network LPs give integer solutions: shortest-path integrality relaxed to $x \ge 0$	●
LP relaxation, branch & bound	Standard IP solution method in Winston Ch. 9	◌

Classifying a new model. (1) Flow conservation at nodes? If yes, start in the network family; if not, general LP. (2) Which devices (page 3) were added? If any of them is integer, it sits below the TU line. “Scheduling” appears three times: **shift** scheduling (covering LP), **machine** scheduling (sequencing IP), and **dispatch** scheduling (shipment consolidation).